

Siyanda Matika

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Data and AI Lead in Johannesburg, with 8 years of experience across banking, capital markets, insurance and research. My work spans the full range of data: large structured datasets in SQL, Spark and Hadoop, ETL pipelines, statistical analysis, and unstructured text, image and video data, through to machine learning and LLM systems in production. In research, I collaborated on data pipelines for petabyte-scale radio astronomy data on the path to the Square Kilometre Array (SKA), as part of international science collaborations.

EXPERIENCE

Manager: AI & Data Science · Discovery Insure

Aug 2024 – present

- Lead a small data science team working on conversational AI, driver-coaching personalisation and the ML platform behind them.
- Building a customer virtual agent for WhatsApp and web: retrieval over product documentation, custom guardrail models and drift monitoring.
- Fine-tuned an agent-routing classifier with Azure OpenAI Service, which reads the user's intent and routes each conversation to the right agent.
- Took AI systems through privacy and security review: wrote Data Protection Impact Assessments (DPIAs), worked with the security team through multiple penetration tests, and worked on data governance and adversarial testing of chatbots and LLMs.
- Built monitoring on Prometheus and Grafana for model services, with alerts on cost, warnings and errors that automatically trigger pipelines.
- Set up Kubeflow pod infrastructure for building agentic systems: agents run in isolated Kubernetes pods cut off from the wider network, so an agent that executes tools or code can only reach the services it is explicitly allowed to use.
- Co-developed a transformer model on sequential health data with Discovery Health; presented it at the Discovery Actuarial and Data Science Conference, where it placed second.

Machine Learning Engineer · Navigare Securities

May 2022 – Aug 2024

- First ML hire: chose the stack, set the roadmap and mentored the engineers who joined.
- Ran the GPU infrastructure for model deployments: served and fine-tuned open-weight LLMs (Llama, Mistral, Mixtral), tuned serving for concurrent users, and built a RunPod scheduling tool with run/sleep cycles to keep GPU spend bounded.
- Designed the AWS ML stack: serverless model services on ECS and Fargate, CI/CD pipelines on GitHub Actions, and data pipelines on Spark and EMR with Delta Lake.
- Built retrieval-augmented assistants for capital-markets clients.
- Built Elections Navigator, a public voter-education chatbot for South Africa's 2024 national elections that answered voters' questions about parties, manifestos and the voting process.

Data Scientist · Absa Group

Jan 2020 – May 2022

- Built a customer lifetime value model and a cross-sell recommender for merchant devices, both used by marketing and cross-sell teams.
- Built customer spend forecasting and customer-health monitoring, using Spark and Hadoop for analytics over the bank's customer data.
- Built Docker containers to deploy the monitoring dashboards.
- Worked with marketing and cross-sell stakeholders to turn model results into customer campaigns and value-proposition recommendations, and presented findings to senior stakeholders.

Business Analyst · Absa Group

Jan 2017 – Jan 2018

- Analysed retail banking data for trends and customer segments.
- Worked directly with product, risk and marketing stakeholders, turning the analysis into recommendations for their teams.

SKILLS

Data engineering	SQL, Spark, Hadoop, Databricks, Delta Lake; ETL pipelines over large datasets
Statistics and analytics	Bayesian inference, time series, segmentation, forecasting, recommendation, A/B testing
Unstructured data	text, images and video: extraction, OCR, embeddings and retrieval
Machine learning	model design, training, evaluation and fine-tuning across classical ML, deep learning and transformers; PyTorch, scikit-learn, XGBoost
LLMs	open-weight models (Llama, Mistral and Mixtral, Qwen) and GPT models via Azure OpenAI; retrieval (RAG, hybrid search), fine-tuning (LoRA, Azure OpenAI Service), intent classification and agent routing, reinforcement learning (GRPO), guardrails
Scripting	Python, Bash, SQL
Cloud services	AWS (SageMaker, ECS, Fargate, EMR), Azure (Azure ML, App Service), GCP
GPU infrastructure and deployment	Linux GPU servers (RunPod), open-weight model serving with vLLM, concurrency tuning, Docker
MLOps and LLMOps	deployment and data pipelines, CI/CD (GitHub Actions, Azure DevOps), Kubeflow on Kubernetes, network-isolated pods for agent workloads
Monitoring	Prometheus, Grafana, model and drift monitoring, alerts that trigger pipelines on cost, warnings and errors
Identity and access	IAM on AWS and Azure, Amazon Cognito, Azure App Service Authentication (Easy Auth)
Security and governance	Data Protection Impact Assessments (DPIAs), penetration-test reviews with security teams, adversarial testing of chatbots and LLMs, data governance
Services	FastAPI, Flask

EDUCATION

PhD research in Astrophysics, not completed · Rhodes University

2018 – 2019

Research on the Epoch of Reionization with the Hydrogen Epoch of Reionization Array (HERA), a precursor to the Square Kilometre Array (SKA).

- Collaborated on pipelines to process and analyse petabyte-scale radio interferometer data, as part of the international HERA collaboration.
- Applied Bayesian inference and Gaussian process regression to separate the faint cosmological signal from foregrounds.

Co-author of:

- Foreground modelling via Gaussian process regression: an application to HERA data, *Monthly Notices of the Royal Astronomical Society* 495(3), 2813–2826, 2020
- Detection of cosmic structures using the bispectrum phase. II. First results from application to cosmic reionization using the Hydrogen Epoch of Reionization Array, *Physical Review D* 102, 022002, 2020

MSc Astrophysics and Space Science · University of Cape Town and University of the Western Cape

2015 – 2017

Joint UCT–UWC degree through the National Astrophysics and Space Science Programme (NASSP). Thesis: *Detecting Galaxies in the Epoch of Reionization*, supervised by Mario Santos: a Bayesian framework for fitting the 21-cm absorption feature in radio galaxy spectra, to measure the redshift of galaxies in the Epoch of Reionization. Also ran SKA1-LOW noise simulations and large-scale early-universe simulations in C++.

BSc Honours Astrophysics and Space Science · University of Cape Town

2014

BSc Physics · University of the Western Cape

2011 – 2013